

CPAN 144 • GROUP PROJECT • PHASE 3

CinemaScope

A Next.js 14 movie discovery app — live TMDB data, optimized, and production-ready.

AGENDA

What we'll cover

01

Project Objective

What CinemaScope is and why we built it

02

Main Features

Core functionality across home, browse, details, watchlist

03

Architecture

Next.js App Router + TypeScript + TMDB

04

Performance

Code splitting, caching, images, debouncing

05

SEO & Metadata

Titles, Open Graph, and dynamic link previews

06

Challenges

What broke, and how we solved it

07

Live Demo

Walkthrough of the running application

Give users a fast, beautiful way to discover movies.

The goal

Turn the Phase 2 UI shell into a production-grade web app: real API data, fast loads, strong SEO, and a polished UX on every route.

What Phase 3 delivers

- Live TMDB API integration (no more mock data)
- Performance passes — bundle, images, caching, search
- Route-level loading / error / 404 states
- SEO metadata + dynamic Open Graph per movie

5

Application routes

8

TMDB endpoints wired

350ms

Search debounce

1hr

Catalog cache window

Five routes plus a mobile-first layout.

Home

Trending hero + Now Playing, Popular, Top Rated, Upcoming carousels.

Browse

Filter by genre, year, rating, and sort. Debounced live search.

Movie Details

Backdrop hero, synopsis, metadata, one-tap add-to-watchlist.

Watchlist

Persisted via React Context + localStorage. Survives refresh.

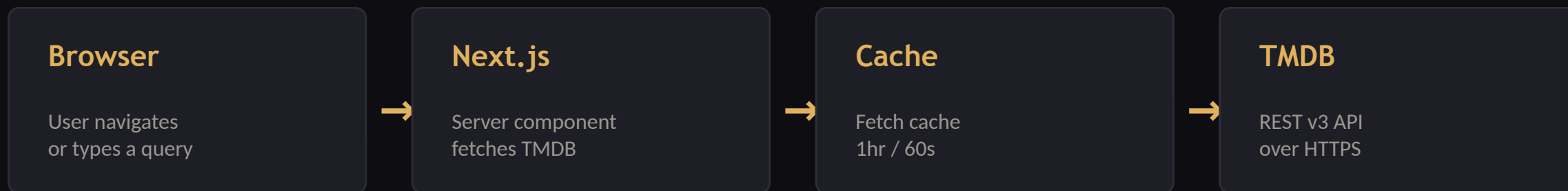
About

Project summary, tech stack, Phase 3 additions.

Responsive

Mobile-first layouts. Works cleanly from 375px to desktop.

Server-first with targeted client interactivity.



Stack

Next.js 14

App Router, RSC

TypeScript

Strict mode

Tailwind

Utility-first CSS

TMDB v3

Movie data + images

Every interactive route is split: a server page.tsx owns metadata + data fetching; a page.client.tsx owns UI state. This pattern unlocks SEO without giving up React state.

Every box on the rubric — ticked.

Code splitting

next/dynamic on Hero + MovieCarousel shrinks initial JS on /

next/image

AVIF + WebP, remote pattern for image.tmdb.org, priority on Hero

Fetch caching

revalidate: 3600 for catalogs, 60 for search — Next.js built-in

Debounced search

350ms useDebounced hook prevents per-keystroke network calls

memo(MovieCard)

Grid + carousel skip re-renders when props are unchanged

Parallel fetches

Promise.all across 5 catalog endpoints on the home route

Link previews that actually look like a product.

What we added

- Root metadata with title template %s | CinemaScope
- openGraph tags (title, description, images, site name)
- twitter card set to summary_large_image
- Keywords, description, and canonical URL
- generateMetadata on /movies/[id] — dynamic per title

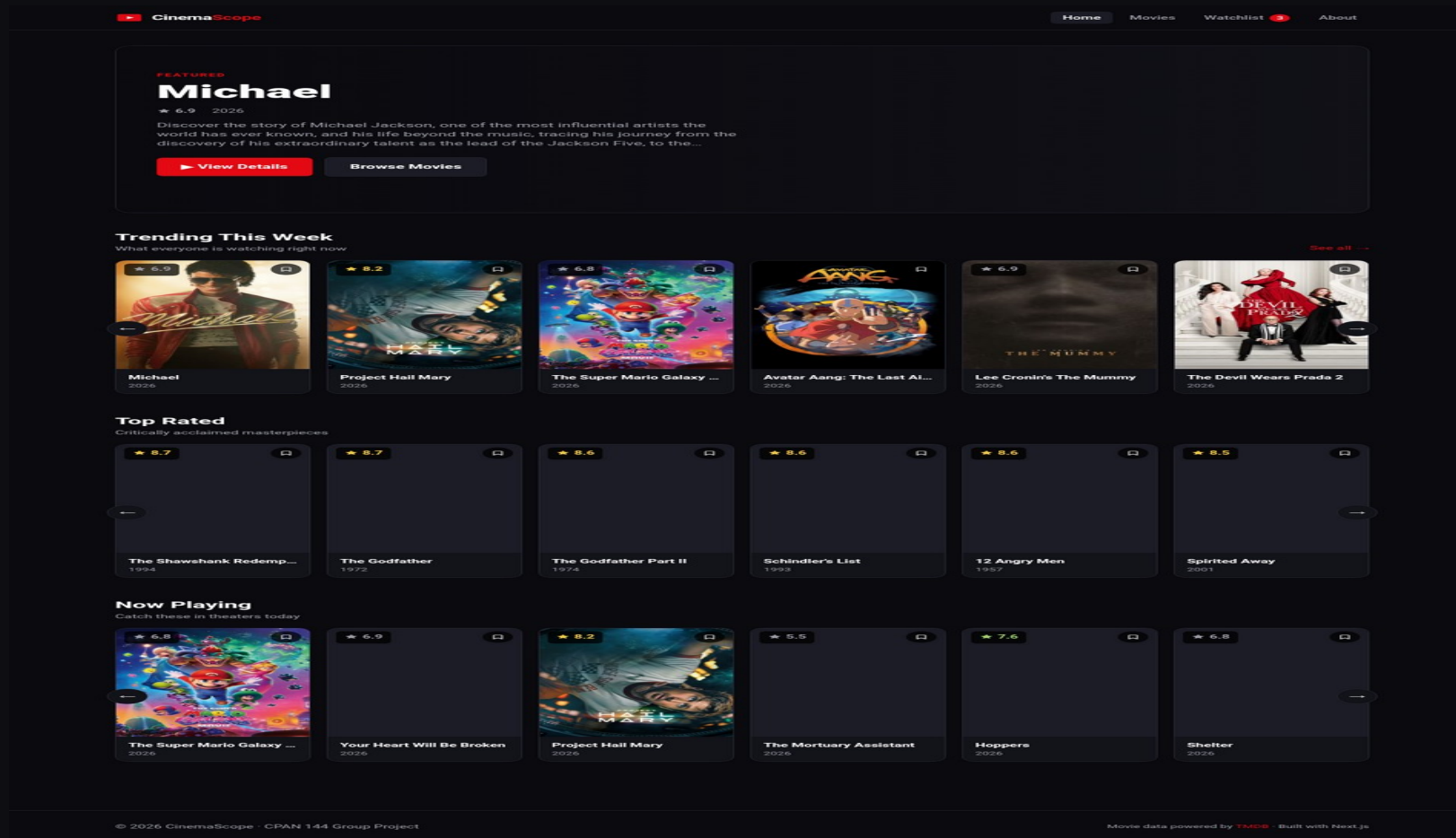
app/movies/[id]/page.tsx

```
export async function generateMetadata({
  params: { id: string }
}) {
  const movie = await getMovie(params.id);
  return {
    title: movie.title,
    description: movie.overview,
    openGraph: {
      title: movie.title,
      images: [tmdbImage(movie.backdrop_path)],
    },
  };
}
```

Six things broke. Six lessons learned.

Challenge	Solution
Client components can't export metadata	Split every page: server page.tsx + client page.client.tsx
TMDB rate limits during fast typing	350ms debounce + 60s fetch cache on /search/movie
next/image blocked external CDN	images.remotePatterns for image.tmdb.org; AVIF + WebP on
Carousels shipped JS on every route	Converted Hero + MovieCarousel to next/dynamic
Hardcoded genre list drifted from API	FilterBar now fetches /genre/movie/list on mount
No UX for slow networks / 404s	Added loading.tsx, error.tsx, not-found.tsx per route

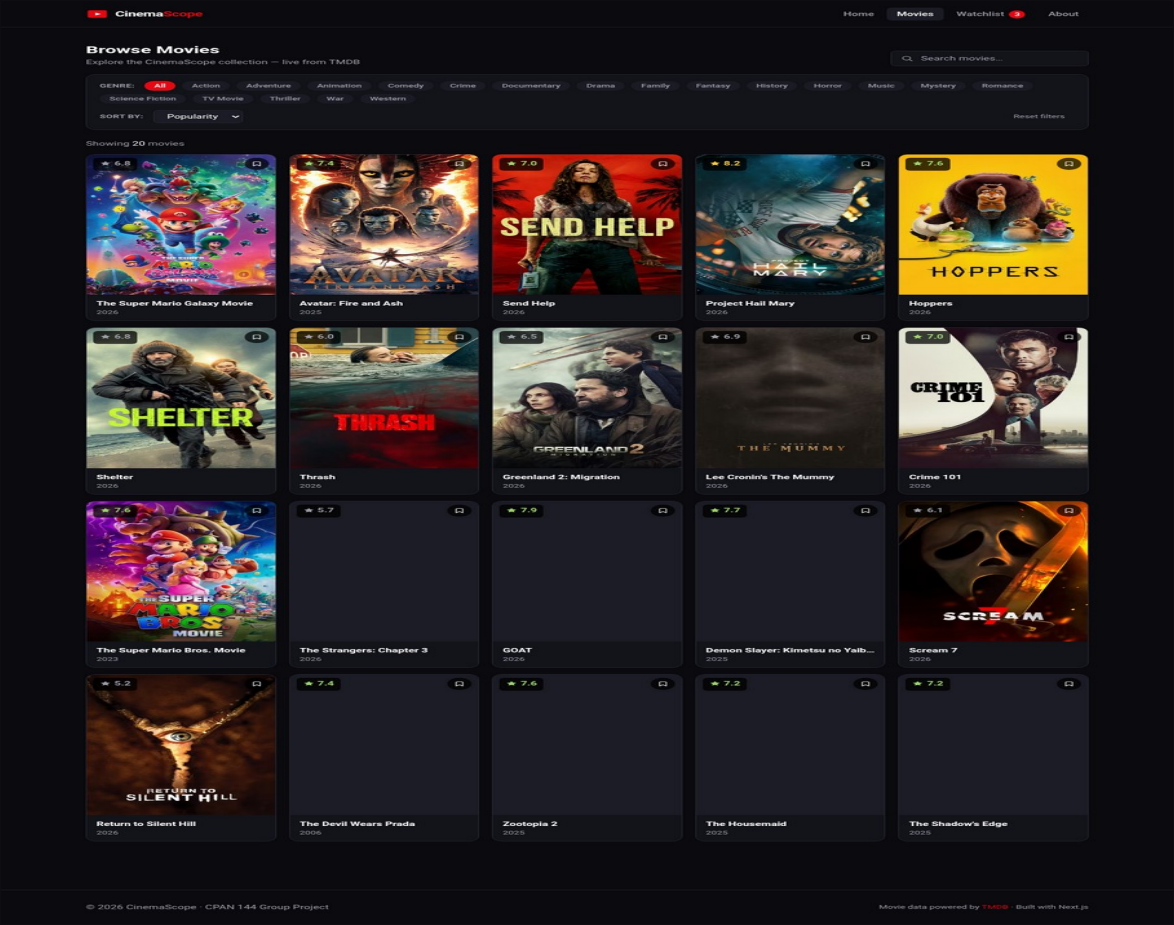
Trending hero + four live carousels.



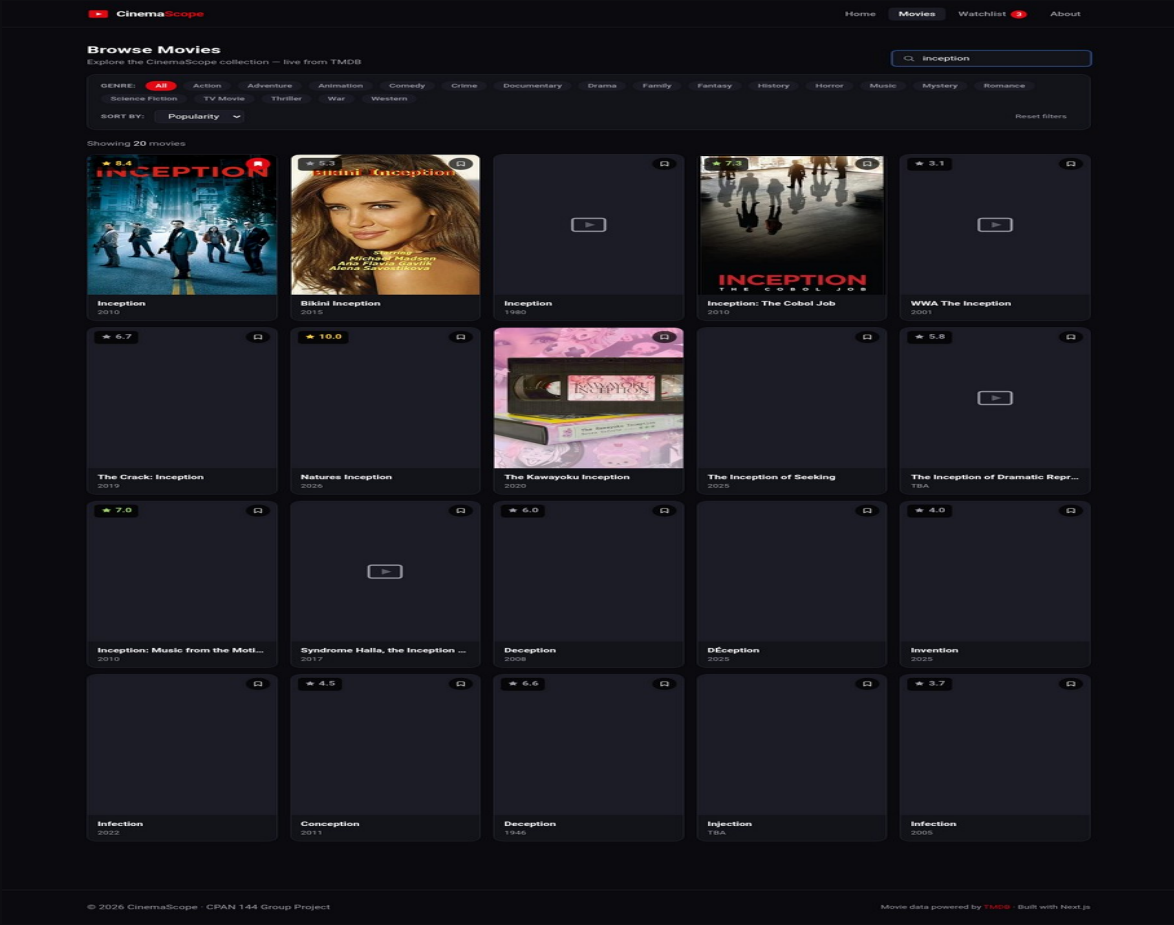
What you're seeing

- Hero loaded with priority for LCP
- Four carousels, one Promise.all
- Posters served as AVIF / WebP
- Dynamic imports on the carousels

Filters and debounced search over live TMDB.

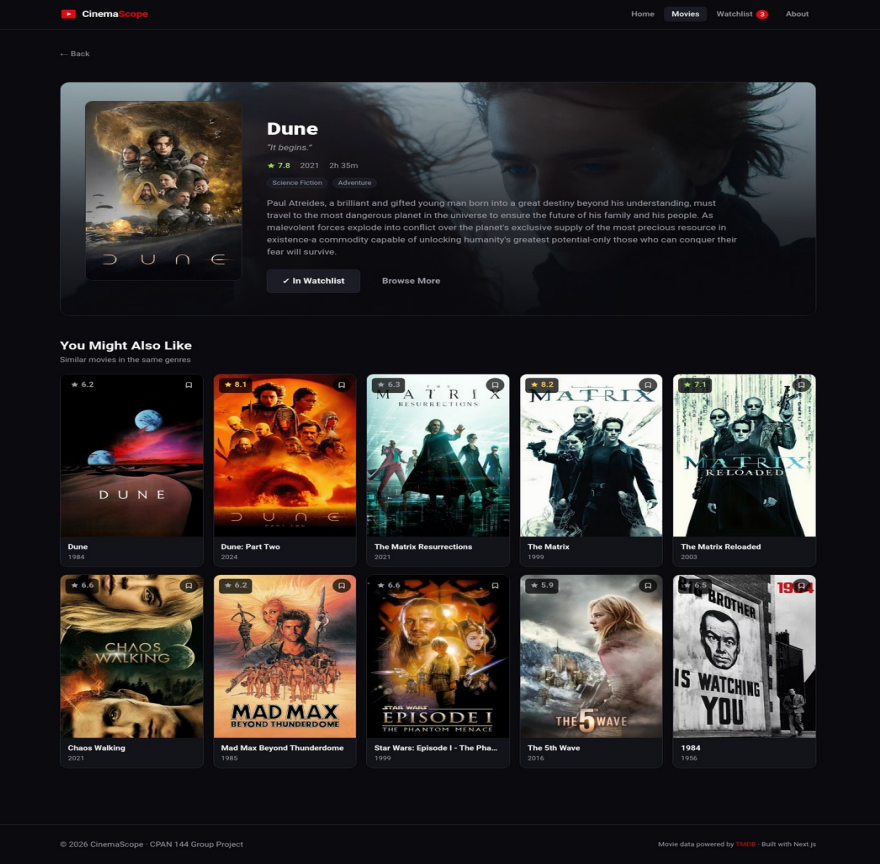


Browse with filters

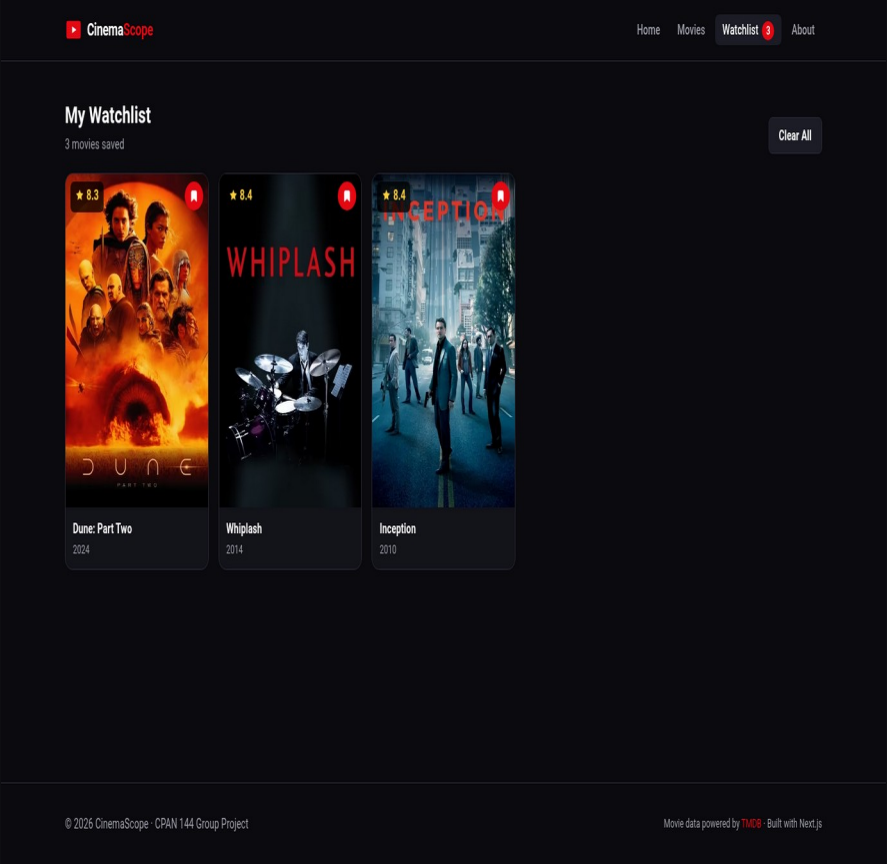


Debounced live search

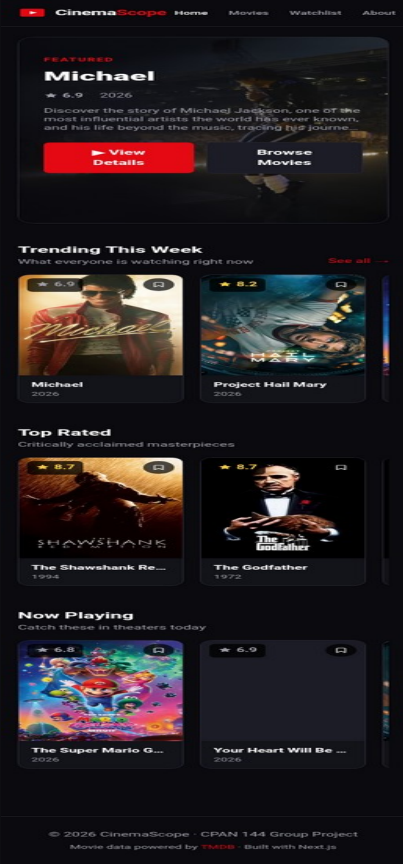
Polished routes across every screen size.



Movie details



Watchlist (persisted)



Mobile • 375px

THANK YOU

Questions?

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Repository

github.com/goothey/cinemascope

Data

[themoviedb.org API v3](https://themoviedb.org)